

RAVITEJA MADDELA

MSc Computer Science & Engineering Student | Ex-Software Developer (6+ Yrs Exp)

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PROFESSIONAL SUMMARY

MSc Computer Science student at the Technical University of Denmark (DTU) with 6+ years of industry experience as a Software Developer across healthcare, fintech, and banking domains (GE Healthcare, Swiss Re, American Express). Specializing in Machine Learning systems, Multi-Agent pathfinding, and scalable backend infrastructure using PyTorch, Java, and Python.

EDUCATION

Technical University of Denmark (DTU) 2025 – Present
Master of Science in Computer Science & Engineering (Copenhagen, Denmark)
Key Focus: Machine Learning Systems, Multi-Agent Systems, Algorithm Engineering, Distributed Systems. Student ID: s252744.
RVR & JC College of Engineering (Acharya Nagarjuna University) 2012 – 2016
Bachelor of Technology in Information Technology (India)

TECHNICAL SKILLS

AI & ML Systems: PyTorch, QLoRA, Transformers, RAG, Mixture-of-Experts (MoE), BioClinicalBERT, FAISS
Backend & Web: Python, Java, FastAPI, Spring Boot, REST APIs, React, Tailwind CSS, Gradio
Cloud & Tools: Docker, Git, Linux, AWS, Kaggle, HuggingFace

FEATURED ENGINEERING & RESEARCH PROJECTS

Sparse MoE-QLoRA for Danish Legal QA PyTorch MoE QLoRA Personal Research Project
Engineered a 4-bit Quantized Low-Rank Adaptation (QLoRA) pipeline to fine-tune a 7B LLM for Danish legal domain QA.
Implemented centroid-based Sparse Mixture-of-Experts (MoE) routing using E5 embeddings paired with FAISS RAG retrieval.
Achieved 6.88 tokens/sec generation throughput and sub-5ms in-place VRAM adapter hot-swapping on a single Kaggle T4 GPU.
Multi-Agent Coordination in Hospital Domain Java CBS A* Search DTU 02285 Course Project
Developed a Java multi-agent path finding (MAPF) solver for autonomous robot agents in the Mavis Hospital Domain grid world.
Implemented Hungarian matching for box-goal allocation, fast greedy priority orderings, and 3-phase box staging.
Built a Conflict-Based Search (CBS) repair engine with CAT bias, opportunistic bypass, Lemma-1 jump points, and segment replanning.
Multi-Modal Clinical Phenotyping System PyTorch FastAPI FHIR R4 Healthcare AI Project
Built a dual-encoder PyTorch pipeline pairing BioClinicalBERT for unstructured clinical notes with a custom Masked Lab-Test Transformer (MLTM) for numerical lab features on MIMIC-IV data (3,397 patient records, 25 conditions, 0.88 AUROC).
Developed a FastAPI inference service with SHAP lab importance, BERT attention scores, FHIR R4 bundle adapters, and Gradio web interface.

PROFESSIONAL EXPERIENCE

Software Developer Jan 2023 – Dec 2023
Infinite Computer Solutions (Client: GE Healthcare) (Bengaluru, India)
Engineered internal license management software for hospital equipment (MRI and CT scanners), enabling staff to easily activate, update, and manage feature licenses without manual intervention.
Streamlined automated installation and upgrade tooling, reducing system downtime and manual overhead for active hospital scanners.
Collaborated across cross-functional engineering teams to ensure on-time delivery of annual software feature upgrades and security patches.
Software Developer Feb 2019 – Jul 2022
Tech Mahindra (Client: Swiss Re) (Hyderabad, India / Zurich, Switzerland)
Developed SwiftMX, a transaction conversion tool mapping insurance business team requests into automated bond and security purchase orders via third-party financial tools (Apex).
Built testing and data migration utilities for TRAX application upgrades, verifying config transfer success rates to guarantee zero downtime.
Implemented custom encryption middleware routing TRAX payment files (salary and insurance disbursements) through multi-system networks (Swift) securely to banking institutions.

Junior Software Engineer

Sep 2016 – Jan 2019

Tata Consultancy Services (Client: American Express) *(Hyderabad, India)*

- Developed online PDF credit statement generation features for American Express Mexico internet banking, replacing inaccessible XML formats.
- Participated in full software lifecycle: requirements analysis, component building, integration testing, and production deployment.

CERTIFICATIONS & LANGUAGES

Certifications: Python Institute Certified Programmer | Lean Six Sigma Yellow Belt | TOEFL iBT (Score: 94/120)

Languages: English (Fluent), Telugu (Native), Danish (Prøve i Dansk 2 / PD2)